2022春高数下期末试题（回忆版）

1、Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) The interval of convergence for the power series $\sum_{n=1}^{\infty} \frac{x^n}{n3^n}$ is

(A) $[-\frac{1}{3}, \frac{1}{3}]$ (B) $[-\frac{1}{3}, \frac{1}{3})$ (C) $[-3, 3]$ (D) $[-3, 3)$

(2) Let $f(x,y) = \begin{cases} y^2 \sin\frac{1}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$. Which of the following statement is wrong?

(A) $f(x,y)$ is continuous at $(0,0)$.

(B) $f_x(0,0)$ exists.

(C) $f_y(0,0)$ exists.

(D) $f_x(x,y)$ is continuous at $(0,0)$.

(3) If $f(x,y)$ has partial derivatives at (x_0, y_0), then

(A) $f(x,y)$ is bounded around (x_0, y_0).

(B) $f(x,y)$ is continuous around (x_0, y_0).

(C) $f(x, y_0)$ is continuous at x_0, $f(x_0, y)$ is continuous at y_0.

(D) $f(x,y)$ is continuous at (x_0, y_0).

(4) Let a be a constant. Then the series $\sum_{n=1}^{\infty} \left(\frac{\sin(an)}{n^2} + \frac{(-1)^n}{n+1}\right)$.

(A) converges absolutely.

(B) converges conditionally.

(C) diverges.

(D) the convergence depends on the value of a.

(5) $\int_0^1 \int_y^1 \frac{\cos x}{x} \, dx dy =$

(A) $\cos 1$ (B) $\sin 1$ (C) $1 - \cos 1$ (D) $1 - \sin 1$.

2、Please fill in the blank for the questions below.

(1) If the function $z = z(x,y)$ is determined by $x^2 - 2y^2 + z^2 - 4x + 2z - 5 = 0$, then $\frac{\partial z}{\partial y}\Big|_{(5,2,2)} =$ _____.

(2) $\lim_{(x,y)\to(0,0)} \frac{\sin(xy^2)}{x^2+y^2} =$ _____.

(3) If the region $D = \{(x,y) \mid x^2 + y^2 \leq 1\}$, then $\iint_D e^{-x^2-y^2} \, dx dy =$ _____.

(4) Let $F = (z + e^{\sin y})\vec{i} + (\cos z - y)\vec{j} + (2z + \ln(1+y^2))\vec{k}$. D is the upper semi-sphere $0 \leq z \leq \sqrt{a^2 - x^2 - y^2}$ $(a \geq 0)$, and S is the boundary of the region D. Then the outward flux of F across S; i.e., $\iint_S F \cdot n \, d\sigma =$ _____.

(5) $\int_C yze^{xz} dx + e^{xz} dy + xye^{xz} dz =$ _____, where C is a path from $(2,1,0)$ to $(0,4,5)$.

3、Find the equation for the plane through the origin parallel to the following lines:

$$l_1 = \begin{cases} x = 1 \\ y = -1 + t \\ z = 2 + t \end{cases}, \quad l_2 = \begin{cases} x = -1 + t \\ y = -2 + 2t \\ z = 1 + t \end{cases}$$

4、Use Taylor series to evaluate $\lim_{n\to\infty} (n^3 \sin\frac{2}{n} - 2n^2)$.

5、In what directions is the directional derivative of $f(x,y) = xy + y^2$ at $P(3,2)$ equal to zero?

6、Compute $\iint_D xy \, dx dy$, here D is the disk enclosed by the curve $x^2 + y^2 = 2x + 2y$.

(Hint: use subsitution.)

7. Find the centroid of the region $D = \{(x,y,z) \mid \sqrt{x^2+y^2} \leq z \leq \sqrt{1-x^2-y^2}\}$.

8. Calculate the line integral $\oint_C F \cdot dr$, where $F = (y^2 + e^{e^x})i + (xy + \cos y)j + xz\,k$, and C is the curve of intersection of the cylinder $x^2+y^2 = 4y$ and the plane $y=z$, ~~con~~ counterclockwise when viewed from above.

9. Find the absolute maximum and minimum values of the function $f(x,y) = 3x^2 + 4xy$ on the region $R: x^2+y^2 \leq 1$.